



Gen AI and Agentic AI in Production

Week 1
Day 5

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

AWS
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

AWS
Bedrock

Terraform

Github
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

FIRST LOOK AT AWS

A look at what's ahead



First, a word about IAM



Granular security is one of AWS's strengths

Super tiresome and painful, but for good reason



We will start by creating a Root User

Only used for assigning permissions & budgeting



Then we will create our “IAM User”

Called **aiengineer** - will be our workhorse for the course

OK let's do this!!

Next, a refresher on API costs



Your responsibility

AWS has offers for people signing up for first time - often 3 months free

Some AWS activities have costs

This week's project is estimated to cost \$1-\$5 if running for a month; for 1-2 days it should be minimal

Always your choice

We will now go and set up alerts

*AWS has alerts but no caps -
**you are responsible for
monitoring API spend!***

Just to give some intuitions

Introducing Docker

What is it



A box within your box

A lightweight alternative to VMs
that shares the host's OS

Why



Isolated, portable

Run the same software
everywhere

Dockerfile

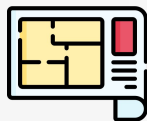


Recipe

A text file with instructions for
installing & configuring



Image

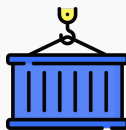


Snapshot

Created by building a Dockerfile
and ready to be made live



Container



A live instance

A running isolated environment
created from an image

Your first 3 AWS services

We will use many more next week..

App Runner



Deploy containers

Simplest AWS way to deploy containerized web applications

ECR



Registry

Like github but for containers on AWS

CloudWatch



Monitoring

Collects logs from all your AWS services

We will package the Healthcare App into a Docker container, test it locally, then deploy

TO THE LAB!



Quick Recap

1. We've built a frontend using **NextJS** in **React**, with **Typescript**, with **Pages Router** (client-side), **Tailwind**
2. We've built a backend using **FastAPI**
3. We've used **Clerk** for user authentication and for subscription plans
4. We've packaged into a **Docker container** and tested it locally
5. Then we deployed the container to **AWS App Runner!**

oh and yes again..

..it **called an LLM and streamed results...** 😄

ASSIGNMENT: Beef up the app! Make it commercially meaningful. Deploy it to Vercel and to AWS to practice both.

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25%
Complete